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Magnetization of Dikes: A Study of the Paleomagnetism of the Widgiemooltha Dike Suite, Western Australia

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Oriented hand samples have been collected at twelve sites in five members of the Widgiemooltha dike suite of Western Australia. These dikes form a major tectonic lineament striking N 80°E, and they have a Rb : Sr isochron age of 2420 ± 30 m.y. At eleven sites, initially scattered directions became well grouped after magnetic cleaning, with a mean direction of $D = 242°$, $I = -67°$. Further magnetic stability is demonstrated by baked-contact tests, and the cleaned results yield a paleomagnetic pole situated in the southwest Pacific Ocean at 157°E, 8½°S with $A_{95} = 8°$. The mean direction lies close to the dike planes. A previous suggestion that by a suitable physical mechanism dikes tend to be magnetized in their plane is investigated. If it is assumed that dikes record the direction of the geomagnetic field at the time of intrusion, a simple geometrical analysis shows that for vertical dikes there is a 50% chance that this direction will lie within 22° of the dike plane. A compilation of seventy-six examples from the literature shows good agreement with the expectations of the analysis. It is suggested, therefore, that the data available at present provide no reason for supposing that dikes do not become magnetized parallel to the ambient field.

INTRODUCTION

Detailed paleomagnetic studies of well-dated Precambrian formations are essential to our knowledge and understanding of the early history of the geomagnetic field. When sufficient information is available, it should be possible to draw important conclusions about a number of geophysical processes involved in the development of the earth. Such processes will include polar wandering, the distribution and relative movement of ancient shield regions, global expansion or contraction, and the origin and possible growth of the core.

At present, there exists a reasonable number of data for some shield areas (e.g., Africa and Canada, see *McElhinny et al.* [1968]), but only preliminary results are available for the Precambrian of Australia [*Irving and Green*, 1958]. The work of *Chamalaun and Porath* [1968] on Australian ore bodies forms a notable exception, although geochronological control was not available; in fact, the paleomagnetic results were used to infer the age of formation of some of the ore bodies [*Porath*, 1967].

This paper describes the results of a paleomagnetic study of a suite of well-dated basic dikes of lower Proterozoic age from the Western Australian shield.

GEOLOGY

The dikes studied belong to the Widgiemooltha dike suite of the Kalgoorlie-Norseman area (Figure 1). They all possess a characteristic strike that varies from 75° to 80° east of north, and, as far as can be determined, they are all vertical or steeply dipping to the south. All the dikes are basic and they include dolerite, gabbro, norite, pyroxenite, and hypersthenite differentiates [*Horwitz and Sofoulis*, 1965].

In the southern part of the area shown in Figure 1 there are two very prominent dikes about 70 to 150 km long and 1 to 2 km wide. These dikes will be referred to below as the Widgiemooltha dike (the more northerly one) and the Norseman dike. Further north they give way to more frequent but smaller dikes that are poorly exposed. The majority of these dikes cause positive magnetic anomalies, and the Bureau of Mineral Resources aeromagnetic maps suggest that the suite may continue many miles north of Kalgoorlie.

All the Archean formations of this area are cut by these dikes, which are themselves truncated in places by younger granite. *Turek* [1966] has conducted a detailed study of the geochronology of this area. For these dikes he obtains an age of 2420 ± 30 m.y. This age is

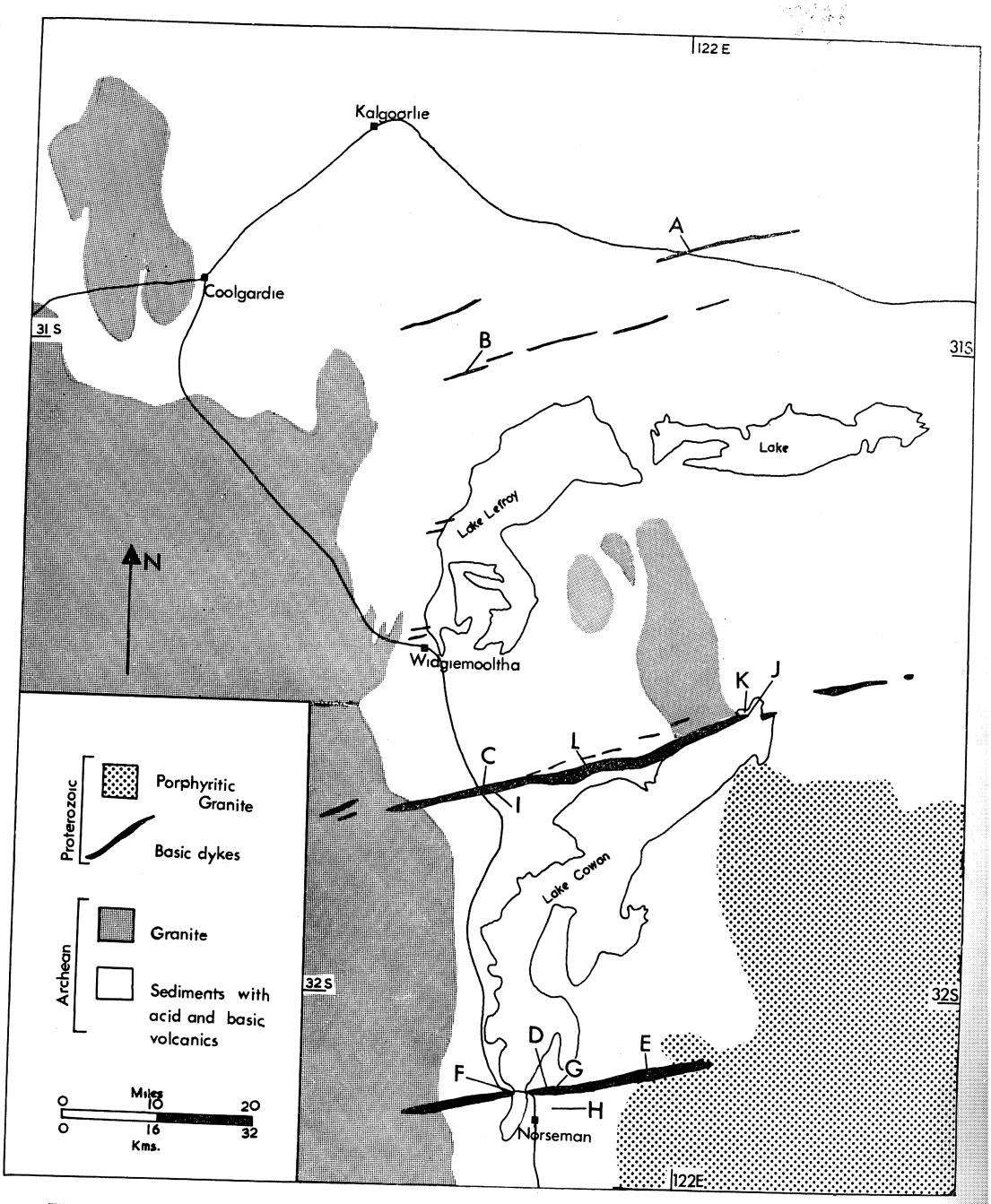
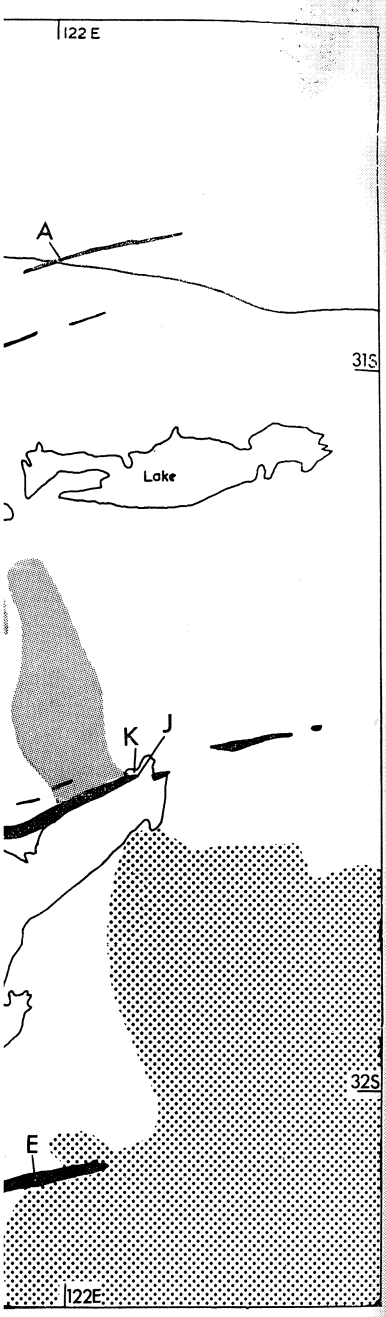


Fig. 1. Geological sketch map of the Kalgoorlie-Norseman area showing the sampling sites.



rea showing the sampling sites.

given by an isochron determined from rubidium-strontium isotopic studies of seven mineral fractions from two of the dikes, using a decay constant of 1.39×10^{-11} yr.$^{-1}$.

Sampling and Measurements

The area forms part of the great inland plateau of Western Australia and lies within the Salinaland physiographic division of *Jutson* [1950]. Despite the generally poor exposure typical of this region, suitable samples were collected at twelve sites situated in five of the dikes. The majority of these samples were restricted to the Norseman and Widgiemooltha dikes, because these two form marked topographic features with more abundant outcrop. Four sites were distributed along 25 km of the Norseman dike and four along 50 km of the Widgiemooltha dike. Three of the remaining four sites were located in three smaller dikes and one in granite adjacent to the Widgiemooltha dike. The sites were distributed between the center and the margins of the dikes and were located in dolerite, gabbro, and pyroxenite rock types. Two to seven hand samples were collected at each of the twelve sites (Figure 1), samples being oriented either by a Brunton compass or a sun-compass clinometer.

From each of the samples a number of cylindrical specimens (2.2 × 2.2 cm) were cut in the laboratory. A total of 110 specimens were obtained from the 43 hand samples.

The direction and intensity of natural remanent magnetization (NRM) of all the specimens were measured on astatic magnetometers, and the directional results are given in Table 1. Only three of the site means had a resultant that could be regarded as nonrandom ($P = 0.05$) by *Watson's* [1956] criterion. The intensities of the NRM are spread over a wide range from 2.0×10^{-2} to 5.0×10^{-6} emu cm^{-3} with an arithmetic mean of 1.69×10^{-3} emu cm^{-3}. The distribution is extremely skew, but, when logarithms are taken, it becomes symmetrical with a mean of 3.68×10^{-4} emu cm^{-3}. A chi-squared test indicates that there is no reason to suppose that the observed distribution of logarithms differs from a true log-normal distribution ($P = 0.05$). Initial susceptibilities, measured on a bridge apparatus similar to that described by *Collinson et al.* [1963], also show a skewed distribution with an arithmetic mean of 4.83×10^{-4} emu cm^{-3} oe^{-1} and a logarithmic mean of 1.76×10^{-4} emu cm^{-3} oe^{-1}.

Alternating-Field Demagnetization

In view of the large scatter of NRM directions all specimens were subjected to alternat-

TABLE 1. Site Mean Directions for the Widgiemooltha Dikes

Site	Site Position Long. °E	Site Position Lat. °S	N	S	NRM D	NRM I	NRM R	NRM k	H_{af}	'Cleaned' Results D	'Cleaned' Results I	'Cleaned' Results R	'Cleaned' Results k	Site Pole Coordinates†
A	122.0	30.9	3	6		*	0.4352		900	239	−67	2.9193	24.8	155.9°E, −6.3°S
B	121.6	31.0	2	6	342	−64½	1.9933	150					*	
C	121.7	31.7	4	8		*	1.0351		750	246	−75½	3.8860	26.3	148.0°E, −17.7°S
D	121.8	32.1	5	15		*	3.0216		750	260½	−69½	4.9758	166	160.6°E, −20.1°S
E	122.0	32.1	3	9		*	0.9031		600	278½	−61	2.7835	9.2	177.4°E, −26.6°S
F	121.7	32.2	3	9	248	−63	2.9868	152	150	243	−66½	2.9831	119	158.1°E, −8.6°S
G	121.8	32.1	3	9		*	2.6125		450	252½	−68	2.9879	166	160.0°E, −14.9°S
H	121.8	32.1	7	14	040	+66	6.9307	86.6	150	051	+73½	6.9301	86.0	145.3°E, −11.1°N
I	121.7	31.7	4	8	189½	−62	3.7262	11.0	450	233½	−68	3.9811	160	152.0°E, −5.3°S
J	122.1	31.6	2	6	203½	−19½	1.7154	3.51	750	224½	−63	1.9196	12.4	152.4°E, +3.9°S
K	122.1	31.6	2	6	213½	−18	1.0531	1.06	450	244½	−60	1.9591	24.5	165.3°E, −3.7°S
L	121.8	31.7	2	5	202½	−64½	1.8283	5.83	900	216	−53	1.9561	22.8	152.6°E, +16.5°S
Mean of eleven site means										242	−67	10.8526	67.9	

Notes:

N, number of hand samples.
S, number of specimens.
D, declination in degrees east of north.
I, inclination in degrees (positive downwards).
R, resultant of *N* unit vectors.
k, precision parameter.
H_{af}, alternating-field used in 'cleaning' (oe).
*, directions are random on *Watson's* [1956] criterion.
†, Positive site pole latitudes are northerly.

ing-field demagnetization using the apparatus and procedure described by *Irving et al.* [1961]. In all but three cases the precision [*Fisher*, 1953] associated with the site mean increased greatly after partial demagnetization, thereby indicating that secondary components had been removed (Table 1). Two of these three exceptions (sites F and H) had well grouped vectors both before and after af treatment, and the third was site B, which possessed NRM vectors parallel to the present field at the sampling locality. During demagnetization the individual specimen vectors for site B moved about in a random fashion and the intensities fluctuated widely; such behavior is typical of unstable magnetization and the results from this site were therefore rejected.

The demagnetization of nine of the remaining eleven sites proceeded normally, but sites J and L required closer attention. No primary magnetization could be isolated from three samples from these two sites. Specimens from these samples either were randomly magnetized or were magnetized many degrees away from the over-all group (determined from the other sites) after demagnetization in the maximum field available (900 oe). These anomalous directions may have been the result of lightning strikes, and the samples were therefore rejected.

Site K is situated in granite about 5 meters from in situ dike rock. The site mean direction from this baked granite after af demagnetization agrees well with the other site means (Figure 2). At site H seven samples were collected from across a 7-meter-wide dolerite dike and from both baked contact zones. Both the NRM and the partially demagnetized vectors were tightly grouped, there being no significant difference between the dike rocks and the baked contact rocks. The agreement between the magnetization of the dike rocks and their baked contacts at these two sites strongly supports the supposition that the stable magnetization of these dikes is the TRM acquired when they originally cooled.

The two large dikes exhibit differentiation and layering, but only at site F was the layering clear enough to permit orientation measurements to be made at the sampling positions. Three samples were taken over about 40 meters across a synclinal axis. The precision parameters (k) before and after correcting the layering to the horizontal are 119 and 18, respectively, and their ratio (6.6) indicates that the dispersion is significantly worse at the 95% confidence level after the correction [*McElhinny*, 1964]. The application of McElhinny's test assumes that the two distributions are Fisherian, and, since N is only 3, this assumption cannot be verified. Nonetheless, the much tighter grouping before correction strongly suggests that the rocks were magnetized in their present position, and therefore the dip of the layering either is original or was impressed above the Curie point of these rocks (500–600°C). A similar conclusion has been given by *McElhinny and Gough* [1963] with regard to the layering of the Great Dyke of Southern Rhodesia.

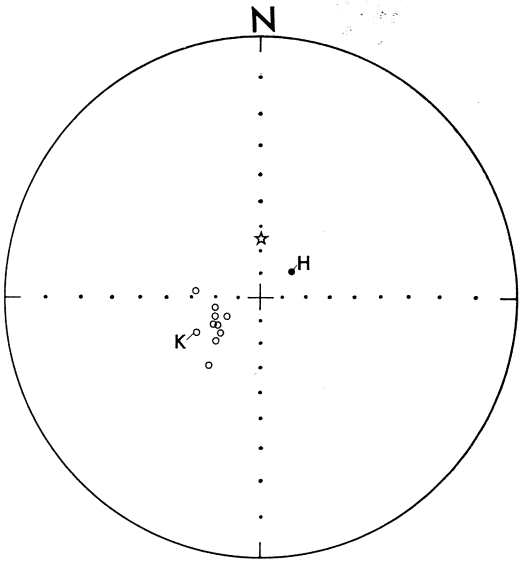
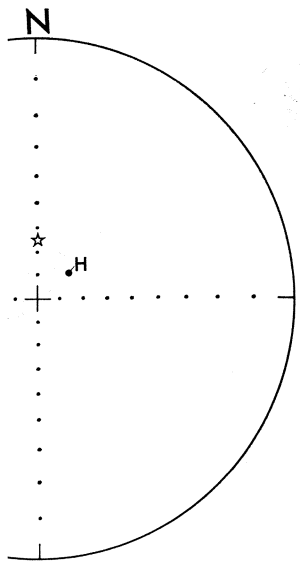


Fig. 2. Site mean directions of magnetization. Stereographic projection, north-seeking directions are plotted. Solid symbols on lower hemisphere; open symbols on upper hemisphere. The star represents the present field (declination = 0).

After partial demagnetization the eleven acceptable site means are closely grouped about a mean direction of $D = 242°$, $I = -67°$. These vectors are plotted in Figure 2, which shows that site H is reversely magnetized with respect to the other sites. Site H is situated 400 meters underground in a gold mine, and samples were taken from a 7-meter-wide dolerite dike. This dike does not outcrop at the surface, but it gives rise to a negative aeromagnetic anomaly that can be traced for 20 km along a strike of



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N 75°E. It is not known whether this small dike is of the same age as the 'normally' magnetized dikes farther north, although there is no geological evidence to suggest otherwise. The close agreement of the direction of magnetization of this dike (irrespective of polarity) with the directions of the 'normally' magnetized dikes indicates at least a penecontemporaneous relationship. This site mean has therefore been included in the calculation of the over-all mean direction of magnetization, for which purpose it was rotated through 180°.

Magnetization of Dikes

The mean remanence vector for the Widgiemooltha dikes lies very close to the plane of the dikes involved, the difference being less than 10°. *Strangway* [1961], in a study of some Canadian Precambrian dikes, observed that their magnetization was always close to the dike plane. In support of his contention that by a suitable physical mechanism dikes tend to be magnetized in their plane, he quoted the results of *Graham* [1953], *Gough* [1956], and *Bruckshaw and Robertson* [1949]. If such a mechanism really operates, it precludes the use of dikes for determining paleomagnetic pole positions. It is therefore important to reassess the situation in the light of the data at present available. To this end, a compilation of paleomagnetic results from dikes has been made from the literature. Only the cases for which the orientation of both the remanence and the dikes themselves are reported could be included; this criterion limited the number of examples to seventy-six. In the compilation, unit weight has been given to dikes, although in 12% of the results insufficient detail is reported, in which case the over-all mean direction of magnetization and an average strike for sub-parallel swarms of dikes have been used. This method yields a more realistic assessment of the data than the method used by Strangway, in which unit weight was given to samples, since it is not biased by variation in sample sizes.

The data have been analyzed in terms of the acute angle between the remanence vector and the dike plane, referred to here as the angle β. In determining β for any dike it was necessary to assume that the dike involved was vertical, unless dip information was reported. In 80% of the compiled results the vertical (or 'steeply' dipping) nature of the dikes is attested by authors' statements or can be inferred from

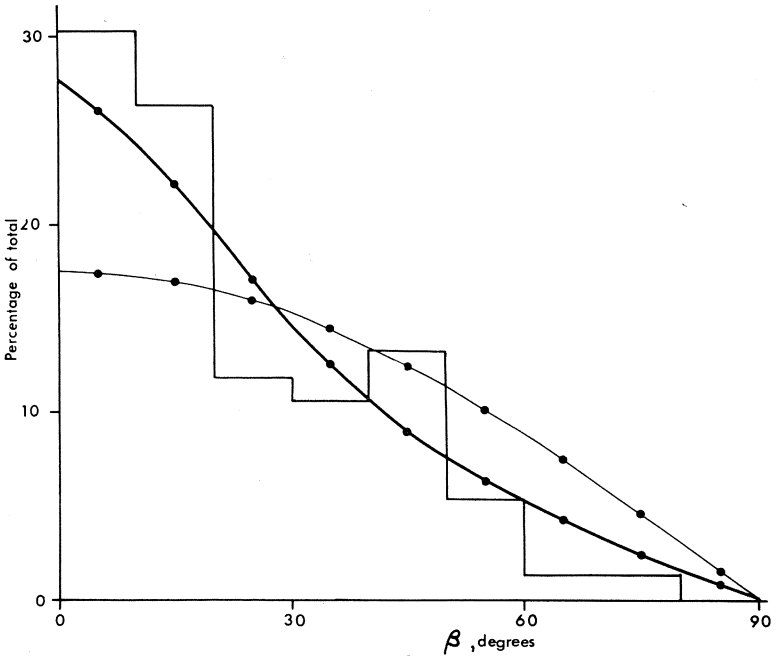


Fig. 3. Histogram of observed values of the angle β. The curves represent the expectations of two theoretical models (see text).

maps showing the linearity of outcrop over extensive distances. The seventy-six observed values of β are plotted as a histogram in Figure 3, which clearly indicates a marked preponderance of low values. Before definite conclusions can be made, however, it is desirable to have a theoretical frequency distribution against which a statistical comparison can be made.

If dikes do acquire remanence parallel to the ambient field, one might expect the remanence vectors of a large number of dikes of different ages to be oriented randomly with respect to the dike planes. Thus, one obvious approach is to calculate the fractions of a *random* set of vectors lying within various angular distances from any plane. This method has been suggested by *Fahrig et al.* [1965], who conclude that 40% of the vectors will lie within 24° of any plane chosen at random. The complete frequency distribution for this model is simply given by the fraction of the total surface area of a sphere (or stereonet) lying between various chosen limits defined by small circles parallel to the plane in question. For example, 1.52% of the surface area of a sphere lies between 80° and 90° from any chosen plane, whereas 17.36% of the surface area lies between 0° and 10°. The expectations of this model, using 10° classes as in the histogram, are represented by the thin curve in Figure 3. It indicates a majority of low values of β, but the curve is much 'flatter' than the observations. The observed and predicted distributions can be compared by means of a χ^2 test which gives $\chi^2 = 21.19^{**}$. This value indicates that the two distributions must be regarded as significantly different ($P = 0.01$). If this curve was a true representation of the theoretical frequency distribution of β, the observed preponderance of low values of β would imply the existence of a distorting mechanism. One would therefore be forced to reject the hypothesis that dikes become magnetized parallel to the ambient field.

The model discussed above assumes random orientations of dikes, whereas in reality there is, almost by definition, a strong tendency for dikes to lie close to the vertical. For this reason the following analysis is preferred.

Consider a dike intruded vertically with strike δ, defined as the acute angle between the dike plane and magnetic north (Figure 4). Let us

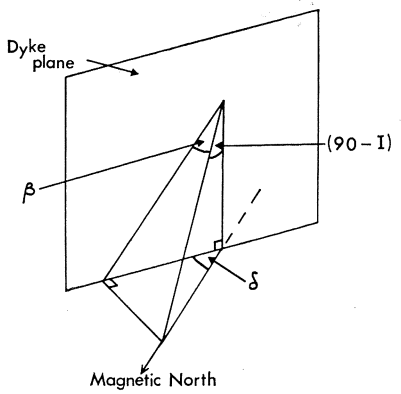


Fig. 4. Definition of the angles δ and β for a vertical dike.

suppose that the remanence of the dike is in fact acquired parallel to the earth's magnetic field at the time of intrusion. A set of east-west dikes have $\delta = 90°$, and β is then merely the complement of the magnetic inclination (I) and will vary with latitude (λ) according to the relation

$$\beta = 90 - I = \cot^{-1}(2 \tan \lambda)$$

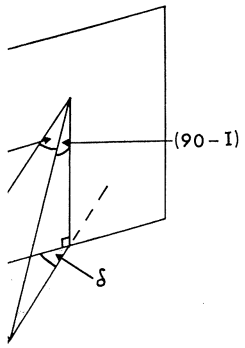
This latitudinal variation in β is represented by the $\delta = 90$ curve in Figure 5. For $\delta \leq 90°$, β is given by the equation (Figure 4)

$$\beta = \sin^{-1}(\cos I \sin \delta)$$

$$= \sin^{-1}\{\cos [\tan^{-1}(2 \tan \lambda)] \sin \delta\}$$

Representative curves for $\delta = 60°$, 30°, and 10° are given in Figure 5, and they serve to illustrate the reasons for expecting a majority of low values of β. It is clear from this diagram that *any* dike that strikes approximately north–south (e.g., $\delta \leq 10°$) will have a small value for β irrespective of its latitude; also *any* dike intruded in high latitudes will have a small β irrespective of its strike. On the other hand, to have a β close to 90°, a dike must be intruded near the equator *and* in an east–west orientation, which is obviously a less frequent occurrence.

To obtain a quantitative estimate of the frequency distribution of β, a diagram has been constructed in which any dike may be represented by a β point determined from its strike and latitude (Figure 6). On such a diagram, curves of constant β may be drawn, and they



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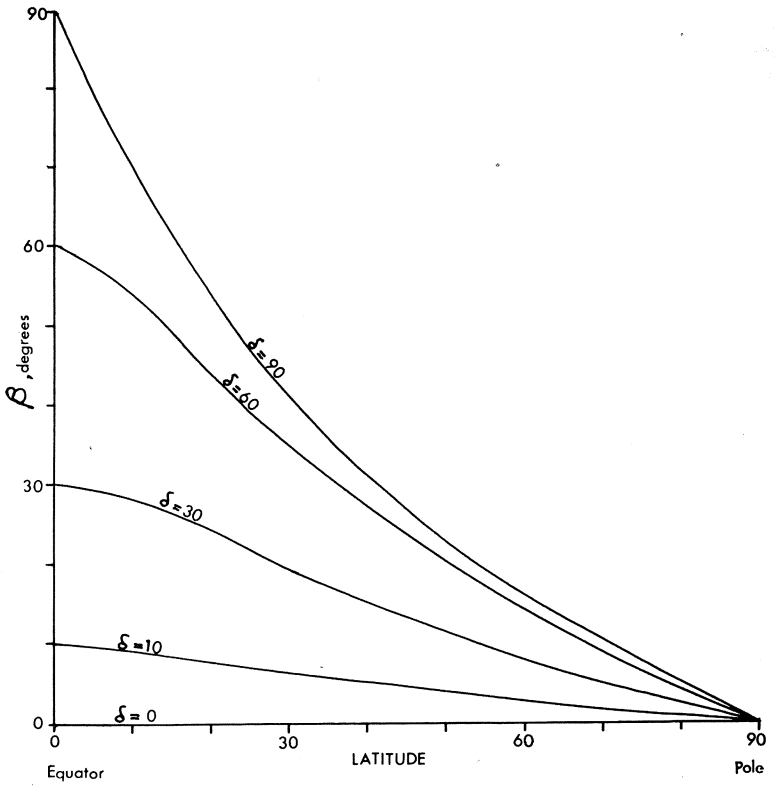


Fig. 5. Variation of the angle β with strike (δ) and latitude (λ).

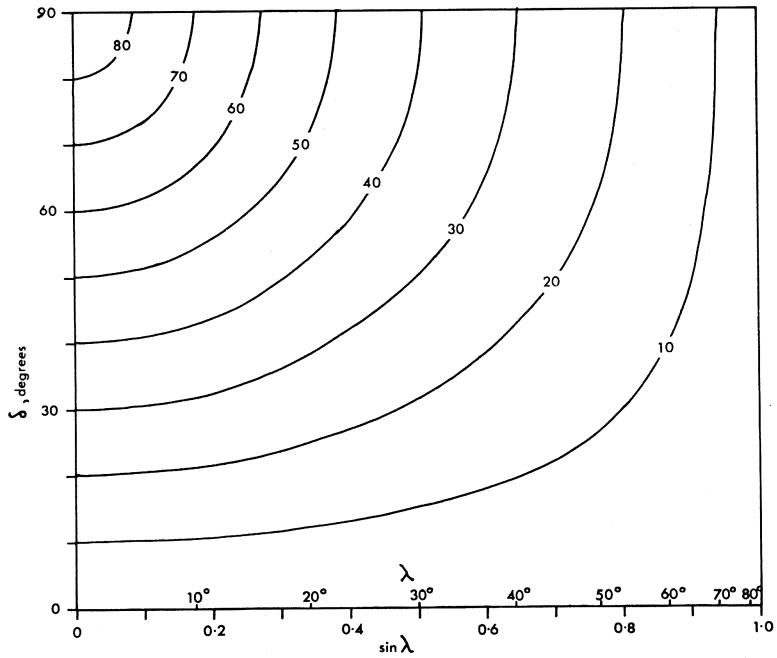


Fig. 6. Diagrammatic representation of the total *sample space* for β for vertical dikes (see text). The numbers on the curves refer to values of β. (δ = strike; λ = latitude.)

divide the total area into zones that represent corresponding ranges of β. If dikes have an equal probability of occurring at any latitude and with any strike, β points will be distributed with uniform density throughout the diagram, and therefore the probability of observing β within a given range will be equal to the fraction of the total area occupied by the zone in question. In a population of randomly oriented dikes, there will be equal probability of any value of strike being observed. A larger proportion of a set of dikes with random geographic distribution will, however, fall in lower latitudes than will fall in polar regions because of the larger surface area involved. Thus, low-latitude ranges must be given extra weight in proportion to the surface area they enclose. The fractional surface area of the globe lying between any two values of λ is equal to the difference between their sines, and hence the simplest method of weighting the abscissa is to plot values of sin λ rather than the values λ itself. In this way one arrives at a diagram (Figure 6) that represents the total *sample space* for vertical dikes and throughout which β points are distributed with uniform density.

For comparison with the histogram (Figure 3), the areas within 10° classes of β have been found by square-counting on a large-scale version of Figure 6. The error involved in square-counting is difficult to determine, but it is no more than 2 or 3%, which is quite adequate for this purpose.

This more realistic model predicts that the probability of observing $\beta \leq 22°$ is 0.5. The complete frequency distribution is given by the heavy curve in Figure 3. It is at once apparent that this curve is in better agreement with the data than the curve predicted by the previous model. A chi-squared test gives $\chi^2 = 6.43$, which is considerably smaller than the $P = 0.05$ significance value of 12.59. In the calculation of χ^2, the number of classes used is only seven since the classes 60°–70°, 70°–80°, and 80°–90° have been combined in accordance with the accepted method of dealing with classes that have low expected frequencies. The low value of χ^2 obtained indicates that there is no reason for doubting that the assumed model is applicable to the observed distribution of β. In other words, the data available at present give no reason to suppose that dikes do not become magnetized parallel to the ambient field at their time of emplacement. This analysis cannot rule out the action of physical mechanisms such as proposed by *Strangway* [1961], but it removes the *necessity* of invoking such explanations.

In this very simple model the effect of dip has been neglected. If this effect were taken into account, it would be very difficult to actually compare the observed data with the model, since the *original* dip of a dike is very rarely determinable. One would not expect, however, the effect of dip to alter the shape of the frequency curve greatly, unless dikes possess a *preferred* dip of something other than 90°. This is a very unlikely possibility, and in any case 80% of the compiled results come from dikes that are known to be close to the vertical at the present time.

A further test can be made by comparing the paleomagnetic pole deduced from dikes with that deduced from other formations of the same age. An excellent example of this comparison is the agreement between the mean Triassic pole for North America [*Beck*, 1965] and the pole reported by *Larochelle and Wanless* [1966] for a large Triassic dike in Nova Scotia (angular difference = 7°). Other good examples are the Tertiary Lundy dikes [*Blundell*, 1957] and the mean Tertiary pole for Britain [*Irving*, 1964, entry 11.026] (6°), the Ferrar dolerites and the Basement dikes that were reheated in Ferrar dolerite times [*Bull et al.*, 1962] (5°), and the general agreement between poles for dikes and poles for flows within the Serra Geral formation [*Creer*, 1962*a*, Figure 4].

A third line of evidence is provided by baked contact studies, since any distorting mechanism occurring within a dike is unlikely to operate also in the country rock. Agreement between dike and baked contact directions in the present study provides evidence against the action of such a mechanism. A number of similar examples exist within the literature, and they have been included in compilations by *Wilson* [1962] and *Irving* [1964, chapter 7 and p. 83].

Thus, at present, there appears to be no general objection to the reliability of paleomagnetic results from dikes.

Paleomagnetic Pole

The over-all mean magnetic vector of the Widgiemooltha dikes is given in Table 1, and it

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ie observed data with the
iginal dip of a dike is very
. One would not expect, how-
lip to alter the shape of the
eatly, unless dikes possess a
nething other than 90°. This
possibility, and in any case
led results come from dikes
be close to the vertical at

n be made by comparing the
deduced from dikes with
other formations of the same
example of this comparison
tween the mean Triassic pole
[*Beck*, 1965] and the pole
elle and Wanless [1966] for
ke in Nova Scotia (angular
Other good examples are the
es [*Blundell*, 1957] and the
e for Britain [*Irving*, 1964,
the Ferrar dolerites and the
at were reheated in Ferrar
l et al., 1962] (5°), and the
between poles for dikes and
hin the Serra Geral forma-
Figure 4].
idence is provided by baked
e any distorting mechanism
dike is unlikely to operate
y rock. Agreement between
tact directions in the present
dence against the action of
. A number of similar ex-
the literature, and they have
mpilations by *Wilson* [1962]
hapter 7 and p. 83].
there appears to be no gen-
he reliability of paleomag-
kes.

MAGNETIC POLE

an magnetic vector of the
s is given in Table 1, and it

corresponds to a paleomagnetic pole in the southwest Pacific Ocean at 157°E, 8°S (dp = 7½°, dm = 9°) on the assumption of a geocentric dipole field. Alternatively, the eleven site poles may be combined to give a mean pole that then lies at 157°E, 8½°S with $R = 10.6785$, $K = 31.1$, and $A_{95} = 8°$.

The sampling sites are widespread both along and across the dikes and are located in three rock types in four dikes (one rejected). Despite this widespread distribution of sites, it is not known whether the time they cover is greater than the 10^4 years believed necessary to average out the disturbing effect of secular variation. The high values of the precision parameter (k = 67.9) for the site mean directions may be interpreted as suggesting that the time involved is small. It is interesting to note, however, that detailed studies on other Precambrian igneous bodies with comparable magnetic inclinations have yielded similar high precisions [e.g., see *Gough et al.*, 1964, Table 1]. It has been common practice in the past to quote the angular dispersion of a Fisherian distribution, $\Delta = 81k^{-1/2}$ degrees, as a measure of the paleosecular variation. For the Widgiemooltha dikes $\Delta = 9.8°$, and an F-test [*Gough et al.*, 1964] gives 95% confidence limits of $6.8° \leq \Delta \leq 14.3°$. Thus, the angular dispersion for this paleolatitude (50°) is not significantly different from that of the present earth's field [*Creer*, 1962*b*], and it falls well within the scatter of actual paleomagnetic observations [*Irving*, 1964, Figure 6.33]. This may be taken to suggest that the time covered by the present collection is, in fact, sufficient to average out secular variation. *Chamalaun* [1968] has suggested, however, that such comparisons be treated with caution since the dispersion caused by the non-dipole field may be somewhat greater than the previously accepted values [*Creer*, 1962*b*; *Cox*, 1962].

The pole position given above is regarded as reliable, since magnetic stability has been demonstrated by baked contact tests and by magnetic cleaning. The mean vector lies close to the plane of the dikes, but a simple analysis of the geometry of vertical dikes shows that this apparent correlation has no significance.

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